

TEZPUR UNIVERSITY

Spring Semester End Term Examination, 2018*Design and Analysis of Algorithms/Algorithms and Complexity (CS 601/701)*

Full Marks : 60

Duration : 2 hours

*Figures in the right hand indicate marks for the individual question***You may answer as many questions as you like but maximum you get is 60.**

1. Following is a variation of Knapsack problem.? [7]

Input: n items of non-negative weights $w_1, w_2, \dots, w_n$, profits $p_1, p_2, \dots, p_n$, positive integers C and S .

Output (yes/no:) Does there exist a subset of items with total weight at most C that correspond to a profit of S ?

Does the above problem belong to the class NP?

2. Consider a row of n coins of values $v_1, v_2, \dots, v_n$, where n is even. Two players P_1 and P_2 take turns to select coins and P_1 starts first. In each turn, a player selects either the first or last coin from the row, removes it from the row permanently, and receives the value of the coin. The player who selects the maximum amount wins. Design an scheme for P_1 such that P_1 wins. [10]
3. Describe the running time of the following pseudocode in Θ notation in terms of n [7]

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for ( $i = n$ ;  $i \geq 1$ ;  $i = i/2$ )
  for ( $j = 0$ ;  $j \leq i$ ;  $j = j + 1$ )
    sum=sum+j;

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4. Answer True / False and provide a counterexample or a short justification. [8×4]
- Consider a directed graph $G(V, E)$ with s as the source vertex. All the edge weights are positive except the edges going out of s which are negatively weighted. Dijkstra's algorithm will generate the correct output for G .
 - If every CUT of a directed graph G has a single crossing edge, then the single source shortest path problem on G can be solved in $O(|V|)$ time.
 - In the Knapsack problem with k items, we decrease the weight of the objects by 1 and decrease the capacity by k . The total profit will more than the profit in the original situation.
 - If Π_1 is NP-complete and $\Pi_1 <_p \Pi_2$, then Π_2 is NP-complete.

- e) In the matrix chain multiplication problem, if the dimension of the matrices are in non-decreasing order, then multiplying the matrices sequentially from right to left will be the most efficient way to get the final product.
- f) If the number of edges in a directed graph G of n vertices is nC_2 edges and each vertex has atleast one outgoing edge, then G is a connected graph.
- g) Consider a path from s to t in a directed graph G . Let $\overline{G}$ be a graph generated from G by reversing direction of all edges of G . Therefore $s \rightarrow t$ path will now be a $t \rightarrow s$ path in $\overline{G}$. The $t \rightarrow s$ path will be the shortest path from t to s in $\overline{G}$.
- h) If edge weights of a connected weighted graph are not all distinct, the graph must have more than one minimum spanning tree.
5. Suppose you are given the filled up table generated by Matrix chain multiplication algorithm for a given sequence of matrices $M_1 \times M_2 \times \dots \times M_n$. You are not given the dimension of the matrices. You are asked to determine the order in which the mtrices are to multiplied. Is it possible? Why or why not? [10]
6. Draw two graphs G_1 and G_2 (each with 6 vertices and 12 edges) for the following two cases: [5+5]
- a) Kruskal's algorithm calls Find(u,v) 7 times to build the MST of G_1 .
- b) Kruskal's algorithm calls Find(u,v) 10 times to build the MST of G_2 .
7. Apply Bellman-Ford's algorithm for the following graph to find the distances of all the vertices form A . What will be the difference if Dijkstra's Algorithm is applied on the same graph? [10+10]

